# Anju Chhetri

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anju-chhetri

## Education

**Technical University of Munich** *Oct. 2026 - Present*
M.Sc. in AI in Society

**Pulchowk Campus - Institute of Engineering, Tribhuvan University** *Nov. 2019 - Apr. 2024*
B.E in Electronics, Communication and Information

## Publications

- **Chhetri, A.**, Shrestha, P., Rana, R., Philip, S., Gyawali, P., & Bhattarai, B. (2026). **When Confidence Lacks Concepts: Interpretable OOD Detection via Representation Perturbations**. (*Preprint*)
- **Chhetri, A.**, Korhonen, J., Gyawali, P., & Bhattarai, B. (2025). **NERO: Explainable Out-of-Distribution Detection with Neuron-Level Relevance in Gastrointestinal Imaging**. *In International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2025*

## Work Experience And Research

**Co-Operations Lead/Founding Team Member** *Mar. 2026 - Present*
AI Safety Nepal Kathmandu, Nepal

- Coordinating education programs and research to build the AI safety community in Nepal.
- Supporting organizational strategy, operations, and community-building.

**Research Assistant** *Apr. 2024 - Jul. 2026*
Nepal Applied Mathematics and Informatics Institute for research Lalitpur, Nepal
Supervisor: *Dr. Binod Bhattarai*

- Working on research projects focused on **trustworthy machine learning**.
- Developed an **OOD detection method** for pathology segmentation in volumetric medical images, specifically MRI and CT scans, based on **self-supervised learning** with synthetic anomalies, leading to a **5$^{\text{th}}$-rank** in the MICCAI 2024 challenge.

**Undergraduate Research Intern** *Aug. 2023 - Mar. 2024*
Nepal Applied Mathematics and Informatics Institute for research Lalitpur, Nepal
Supervisor: Dr. Binod Bhattarai

- Worked on an **Active Learning framework** to determine the most effective **uncertainty-based sampling technique** to address dataset imbalance in a multi-class object detection task.
- Trained deep learning models and conducted an ablation study for detecting gait abnormalities in warehouse environments.

## Scholarships and Fellowships

1. Master's scholarship (*Konrad Zuse School of Excellence in Learning and Intelligent Systems (ELIZA)*) *Oct.1, 2026*
2. TUM-Need based scholarship (*Technical University of Munich*) *Sep.15, 2026*
3. Nepal Government Undergraduate Study Grant (*Government of Nepal*) *2019-2024*
4. Mahatma Gandhi Scholarship (*Embassy of India*) *2017-2018*

## Teaching Experience

**Fifth Annual Nepal AI School - NAAMII** *27th Dec., 2024 - 6th Jan., 2025*
Teaching Assistant *Certificate*

- Performed live demonstrations of adversarial attacks in image models and LLMs to an audience of 100 students under the supervision Prof. Yash Raj Shrestha (University of Lausanne, Switzerland).
- Assisted in lab sessions on Geometric Deep Learning, Active Learning, and Self-Supervised Learning.

## Selected Academic Projects

**Improving Adversarial Robustness For Preventing Image Manipulation** *Oct. 2023 - Apr. 2024*

Capstone Project *Link to Repo*

- Developed a **JPEG-resistant image immunization pipeline** to prevent text-guided AI manipulation, specifically for text-guided diffusion models.
- Investigated the impact of text prompts on the **success of adversarial attacks** against **text-guided diffusion models**.

**Real-Time Visual Element Generation Using GANs** *Dec. 2022 – Apr. 2023*

Academic Project *Paper* | *Link to Repo*

- Developed a **real-time neural rendering system** for generating realistic grass in video games.
- Experimented with existing architectures, performed result analysis, and designed new architectures that achieved both visual fidelity and low inference time.

## Awards and Competitions

Nepal-US Hackathon - **Category Winner (Innovative)** *March 2026*
Nepali Leaders Network

Eastern European Machine Learning - **Full Scholarship** *July 2025*
*ANNT, Google DeepMind, University of Sarajevo*

Mentee, Women Leaders Mentorship Programme, *EU in Nepal* *Dec. 2024–October 2025*
Mentor: **H.E. Veronique Lorenzo**, EU Ambassador to Nepal, **4% acceptance rate**

Medical OOD Analysis Challenge, **5th Place**, Pixel-level Task *October 2024*
*MICCAI 2024*

Hacking for Humanity 2023, **2nd Runner Up** *February 2023*
*Shequal Foundation*, Nepal's largest all female hackathon

GritFeat AI Health Hackathon, **1st Position** *January 2023*
*GritFeat Solutions*

ABU-Robocon 2022, **2nd Runner Up** *August 2022*
*Asia-Pacific Robot Contest*

3rd Annual Nepal AI School - **Full Scholarship** *December 2021*
*Nepal Applied Mathematics and Informatics Institute for Research*

## Service

### Reviewer

- 29th International Conference on Medical Image Computing and Computer Assisted Intervention (MICCAI 2026)
- Deepfusion Nepal Computer Science Researchers Meetup and Mini-Conference, IEEE Computer Society Pulchowk, 2026

### Student Activities

- Event Manager, IT Club - Pulchowk Campus, 2021-2023
- Student Member, Robotics Club - Pulchowk Campus, 2019-2022

### Volunteering

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| 03/2026 – ongoing | *Co-Operations Lead* **(AI Safety Nepal)**: Talks and workshops on AI and AI safety for general public and schools. |
| 5/2025 – 09/2025 | *Web and Publicity Chair* **(DEMI workshop at MICCAI 2025)**: managing the online presence and public communication. |
| 10/2024 – 12/2024 | *Organizing and Admission Committee* **(Fifth Annual Nepal AI School)**: Managed logistics, including stakeholder coordination, task division, and application evaluation |
| 08/2024 | *AI Trainer, Right to Information* **(Sancharika Samuha, The Asia Foundation, USAID Nepal)**: Supported mid-career journalists in eastern Nepal on enhancing their understanding of AI technology. |
| 11/2023 | *Children in Technology, 2023* **(LOCUS 2023)**: Awareness program on internet safety and cyber law to underserved rural communities in eastern Nepal. |